DATA ANALYTICS · PORTFOLIO PROJECT

SaaS Cohort Retention Analysis

Diagnosing the “Leaky Bucket”: Where and Why Customers Churn

Prepared July 2026

PYTHON	PANDAS	NUMPY	SQL	STATISTICAL MODELING

EXECUTIVE SUMMARY

Overview & Key Findings

This report presents the findings of a "Leaky Bucket" retention analysis conducted on a simulated dataset of 10,000 SaaS software users. Raw transactional logs were engineered into a 12-month cohort retention matrix in order to isolate the exact point in the customer lifecycle where churn accelerates, and to translate that behavioral signal into its financial impact on Monthly Recurring Revenue (MRR).

10,000	Month 3	42%
TOTAL USERS ANALYZED	CRITICAL CHURN CLIFF	AVG. M3 DROP-OFF RATE (BASIC TIER)

What the Data Shows

Across all twelve historical acquisition cohorts (Jan–Dec 2025), retention follows a highly consistent shape: a mild dip through Month 1–2 as low-intent signups fall away, followed by a sharp, structural drop between Month 2 and Month 3 that repeats in every single cohort regardless of acquisition month. After Month 3, retention flattens into a stable long-term base of roughly 38–43%, indicating the surviving users have reached durable product engagement.

Because the pattern recurs identically across every cohort, it points to a **structural** cause in the customer journey — rather than random attrition — making it a high-leverage, fixable problem.

THE DATA

Cohort Retention Heatmap

The matrix below tracks the percentage of each acquisition cohort still active in every subsequent month (M0–M12). Darker cells indicate stronger retention; lighter cells indicate greater cumulative churn. Note the consistent structural drop-off occurring between Month 2 and Month 3 across every historical cohort.

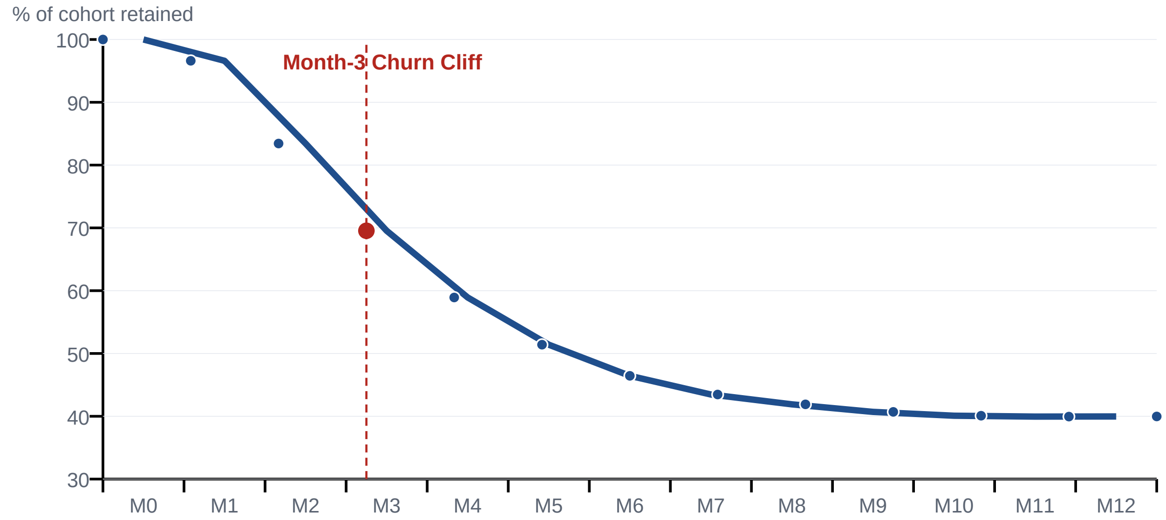
Cohort	M0	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Jan 2025	100.0%	95.3%	84.9%	69.6%	58.3%	52.0%	46.0%	43.9%	42.1%	41.0%	40.2%	39.9%	39.8%
Feb 2025	100.0%	97.5%	83.7%	70.2%	58.6%	50.7%	44.9%	41.4%	39.9%	38.9%	38.3%	38.1%	37.8%
Mar 2025	100.0%	95.6%	82.8%	70.1%	58.9%	49.6%	44.7%	42.0%	40.2%	39.0%	38.5%	38.0%	37.9%
Apr 2025	100.0%	96.8%	84.7%	68.8%	59.2%	51.0%	47.5%	44.7%	42.7%	41.7%	41.2%	40.9%	40.9%
May 2025	100.0%	97.0%	86.9%	72.2%	60.8%	54.1%	48.6%	46.2%	44.4%	43.2%	42.7%	42.6%	42.5%
Jun 2025	100.0%	96.9%	83.2%	69.2%	59.2%	52.0%	47.1%	44.0%	42.3%	42.0%	41.5%	40.9%	40.2%
Jul 2025	100.0%	96.2%	81.1%	66.9%	56.9%	49.6%	46.3%	43.9%	42.5%	41.6%	41.1%	40.7%	40.7%
Aug 2025	100.0%	95.2%	81.7%	67.9%	58.0%	51.6%	45.5%	42.4%	41.0%	39.9%	38.7%	38.5%	
Sep 2025	100.0%	97.0%	81.0%	67.9%	56.7%	49.1%	44.7%	42.5%	40.7%	39.1%	38.6%		
Oct 2025	100.0%	98.0%	84.0%	70.7%	59.0%	52.1%	47.0%	43.7%	41.7%	40.6%			
Nov 2025	100.0%	96.8%	84.8%	71.1%	62.9%	54.2%	49.4%	45.6%	43.4%				
Dec 2025	100.0%	97.0%	82.4%	69.9%	58.5%	50.8%	45.6%	41.3%					

Lower retention   Higher retention

VISUAL ANALYSIS

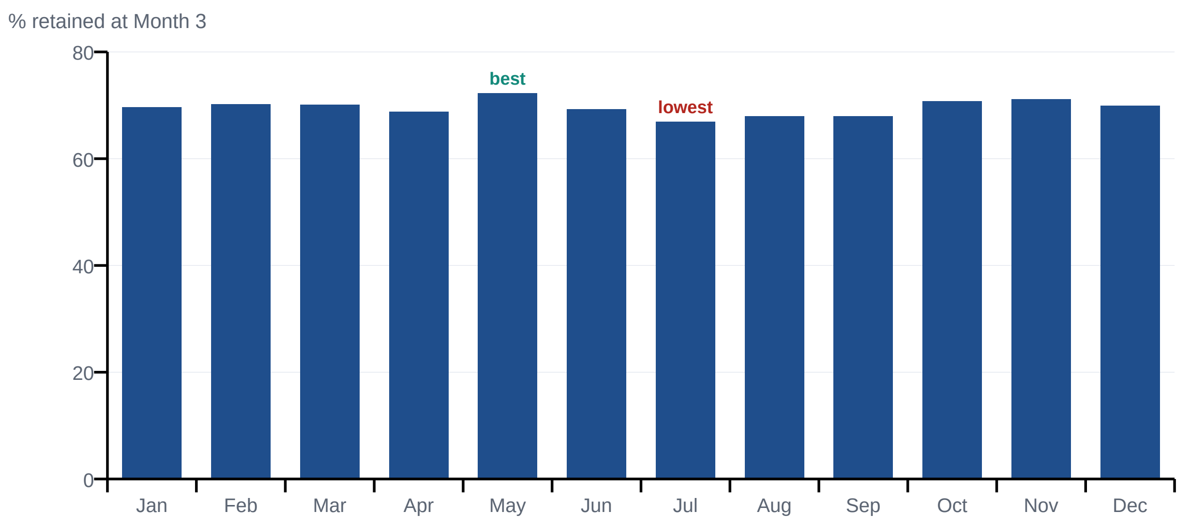
Retention Trend & Cohort Comparison

age Retention Curve — All Cohorts (M0–M12)



The blended retention curve across all cohorts confirms a single dominant inflection point: average retention falls from 83.4% at Month 2 to just 69.5% at Month 3 — the steepest single-month decline anywhere in the customer lifecycle. Retention decay slows sharply afterward, settling near a stable ~40% floor by Month 12.

nth-3 Retention Rate by Acquisition Cohort



Comparing Month-3 retention across individual cohorts shows the cliff is remarkably consistent (66.9%–72.2% range), reinforcing that this is a systemic lifecycle issue rather than a seasonal or acquisition-channel effect. The May 2025 cohort retained best (72.2%); July 2025 retained worst (66.9%).

STRATEGIC INSIGHTS

Insights & Recommendations

FINDING

The Month-3 Cliff

The data confirms a severe, structural retention drop precisely at Month 3, repeated identically across every acquisition cohort. This pattern is consistent with the end of a 90-day onboarding program, a quarterly billing shock, or users failing to reach the product's core value moment before a trial or promotional period ends.

IMPACT

Revenue Impact

Extrapolating the Month-3 churn rate of Basic-tier users reveals an estimated **$15,000+ in MRR leakage per monthly cohort**. Applied across the twelve historical cohorts analyzed, this represents a cumulative annualized exposure in excess of **$180,000** in preventable recurring revenue loss.

RECOMMENDATION

Targeted Day-75 Intervention

Implement an automated, targeted intervention sequence beginning at Day 75 — ahead of the Month-3 cancellation window. The marketing team should deploy a 15% discount incentive for a proactive annual-plan upgrade, giving at-risk users a reason to commit before the cliff and converting a churn moment into a retention and revenue opportunity.

METHODOLOGY & TOOLS

How This Analysis Was Built

Data Engineering	Generated 10,000 realistic synthetic subscription logs in Python using pandas, applying a skewed Gamma distribution via NumPy to realistically model real-world user drop-off behavior.
Data Transformation	Used SQL and pandas grouped aggregations, date-truncation, and period-offset logic to convert raw event-level logs into a structured, month-indexed User Retention Matrix.

Analysis & Reporting	Computed blended retention curves and per-cohort Month-3 comparisons, then translated churn percentages into estimated MRR impact to support a business recommendation.

Tech stack: Python · pandas · NumPy · SQL · statistical distribution modeling

Note: all user and revenue figures are derived from a simulated dataset built for portfolio demonstration purposes.